BIOINTEL

Practical Mini Project Report

Bioinformatics and Information Retrieval

Intelligent Biomedical Information Retrieval and Knowledge Discovery Platform

Submitted by:

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1. Abstract

BioIntel is a full-stack web application developed as an academic mini-project for Bioinformatics and Information Retrieval. It helps users search biomedical research literature and discover connections between biological entities such as genes, diseases, proteins, and therapeutic drugs. Instead of relying only on exact keyword matching, the system uses text preprocessing, an inverted index, TF-IDF with cosine similarity, Okapi BM25, and semantic concept matching to retrieve and rank relevant papers. It also highlights biomedical entities, displays an interactive knowledge graph, and provides analytics and dataset-management features. The project demonstrates how Information Retrieval and biomedical data organization can work together in a practical research-support tool.

2. Introduction

Biomedical research produces a large amount of scientific literature. Finding useful information about a gene, disease, protein, or medicine can take time, especially when researchers use different terms or synonyms for the same concept. Basic keyword search may miss relevant documents or rank longer documents unfairly.

BioIntel addresses this challenge by combining an Information Retrieval pipeline with biomedical entity extraction and knowledge discovery. Users can submit queries such as “BRCA1 mutations associated with breast cancer,” inspect ranked papers, explore detected entities, and view relationships between biological concepts. The application is intended for academic demonstrations and laboratory examinations.

3. Problem Statement and Objectives

Problem Statement

Biomedical literature and biological information are often spread across many documents and data sources. Simple search methods may not handle specialized terminology, related concepts, or document relevance consistently, and they may not explain why a result was ranked highly. A unified platform is needed to search biomedical papers, identify important entities, and explore their relationships.

Objectives

1. Preprocess biomedical queries and research text using domain-aware tokenization and stop-word removal.
2. Build an inverted index to retrieve candidate documents efficiently.
3. Rank research papers using TF-IDF, cosine similarity, Okapi BM25, and semantic concept matching.
4. Combine ranking signals through a configurable hybrid relevance score.
5. Identify and highlight genes, diseases, proteins, and therapeutic drugs in paper text.
6. Visualize biological relationships using an interactive knowledge graph.
7. Provide analytics, search history, and CSV-based dataset import and export.

4. Information Retrieval and NLP Approach

A. Biomedical Text Preprocessing

The system converts text to a consistent form, tokenizes it, and removes common English stop words and academic filler terms. Tokenization is designed to preserve biomedical names and symbols such as BRCA1, TNF-alpha, and anti-PD-1.

B. Inverted Index

An inverted index maps each vocabulary term to a list of documents in which it appears. The postings can include document frequency and term frequency information. This allows the search engine to locate candidate documents without scanning every document in full for each query.

C. TF-IDF and Cosine Similarity

TF-IDF gives greater importance to terms that are frequent in a particular document but less common across the whole collection. Cosine similarity compares the query and document vectors, helping rank papers according to their textual similarity to the query.

D. Okapi BM25

BM25 is a probabilistic ranking method that considers how often query terms appear and normalizes for document length. It helps avoid giving overly high scores to a document merely because its abstract is long.

E. Semantic and Hybrid Ranking

Semantic concept matching connects selected related biomedical terms, such as oncology and cancer, to improve retrieval when wording differs. The hybrid ranking option combines TF-IDF, BM25, and semantic scores using configurable weights. The result inspector explains the component scores and matched terms.

F. Biomedical Entity Extraction

A curated dictionary and regular-expression-based matching identify four entity groups: genes, diseases, proteins, and drugs. The detected entities are highlighted in research text and can be explored through dedicated catalogs. The project description associates these catalogs with resources such as HUGO/HGNC, MeSH, UniProt, and FDA drug information.

5. System Workflow

8. The user enters a biomedical query in the web interface.
9. The query is normalized, tokenized, and filtered to remove stop words.
10. The inverted index retrieves candidate research papers.
11. TF-IDF, BM25, and semantic concept scores are calculated for candidate papers.
12. The system combines ranking scores and sorts the papers by relevance.
13. Biomedical entities are detected and highlighted in titles and abstracts.
14. The user reviews paper details and the “Why this result?” scoring explanation.
15. The user can explore related papers, entity catalogs, analytics, and the knowledge graph.

Suggested place for a manually added figure: Insert a system workflow or architecture diagram here.

6. Technology Stack

Component	Technology	Purpose
Frontend	React 19, TypeScript, Vite	Builds the interactive single-page web interface.
Styling	Tailwind CSS	Provides responsive layout and interface styling.
Backend API	Python, FastAPI, Uvicorn	Handles API requests and backend application logic.
Data validation	Pydantic	Validates request and response data.
Database / ORM	SQLAlchemy with SQLite; PostgreSQL-compatible design	Stores papers and biological entity records.
Information Retrieval	Inverted Index, TF-IDF, cosine similarity, BM25	Finds and ranks relevant research papers.
Entity extraction	Curated dictionaries and regex-based NER	Identifies genes, diseases, proteins, and drugs.
Visualization	Recharts and custom SVG graph	Displays analytics and biological relationships.
Data management	CSV import/export and browser LocalStorage	Supports custom datasets and search-history persistence.

7. Key Features

- Biomedical Literature Search: Searches research papers using domain-specific queries.
- Multiple Ranking Methods: Supports TF-IDF, BM25, semantic matching, and hybrid ranking.
- “Why This Result?” Inspector: Displays matched terms and individual relevance-score components.
- Biomedical Entity Highlighting: Identifies genes, diseases, proteins, and drugs in paper text.
- Entity Explorers: Provides searchable catalogs for the four biological entity categories.
- Interactive Knowledge Graph: Displays relationships such as associated_with, expresses, inhibits, treats, and regulates.
- Related Research: Recommends papers with similar abstracts.
- Analytics Dashboard: Presents publication trends, entity distributions, and frequent terms.
- CSV Dataset Manager: Supports importing and exporting literature data and restoring seed data.
- Search History and Viva Inspector: Allows previous searches to be rerun and provides an inverted-index inspection tool.

8. Performance and Evaluation

The project can be evaluated using standard Information Retrieval measures. Precision at K measures how many of the top K retrieved papers are relevant. Mean Average Precision (MAP) evaluates ranking quality across multiple queries, while Mean Reciprocal Rank (MRR) measures how highly the first relevant result appears. Retrieval latency can be recorded to assess speed.

The ranking methods can be compared using the same set of biomedical queries and manually judged relevance labels. No measured performance values are stated in this report because they were not provided in the supplied project description; numerical results should be added only after testing.

9. Advantages and Applications

- Makes biomedical literature easier to search and explore from one interface.
- Combines classical Information Retrieval methods with biomedical terminology handling.
- Makes ranking more understandable through a visible score breakdown.
- Connects papers with genes, diseases, proteins, and therapeutic drugs.
- Supports teaching and laboratory demonstrations of indexing, ranking, and entity extraction.
- Provides analytics and CSV tools for exploring or updating a curated dataset.

10. Conclusion

BioIntel demonstrates how Information Retrieval and Bioinformatics can be integrated into a single web-based knowledge discovery platform. Its inverted index supports efficient candidate retrieval, while TF-IDF, cosine similarity, BM25, and semantic concept matching help rank relevant papers. Biomedical entity extraction and knowledge graph visualization add context beyond keyword search. Analytics, search history, and dataset management make the platform useful for academic demonstrations and further development. Future work could include larger verified biomedical datasets, evaluation against expert relevance judgments, and integration with trusted live literature sources.

11. References / Project Sources

16. Manning, C. D., Raghavan, P., and Schütze, H. (2008). Introduction to Information Retrieval. Cambridge University Press.
17. Robertson, S. E., and Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in Information Retrieval.
18. National Library of Medicine. PubMed / MEDLINE and Medical Subject Headings (MeSH) resources.
19. HUGO Gene Nomenclature Committee (HGNC) resources for gene nomenclature.
20. UniProt Knowledgebase resources for protein information.
21. BioIntel project description and architecture notes supplied for this mini-project.

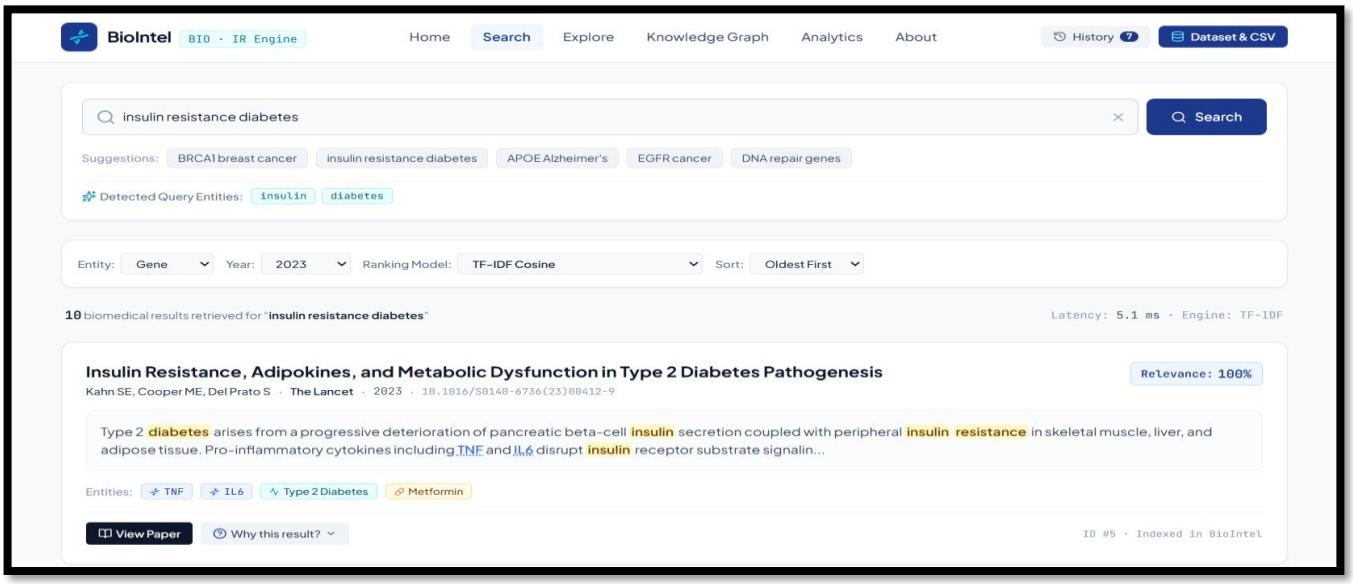
Suggested Images to Add Manually

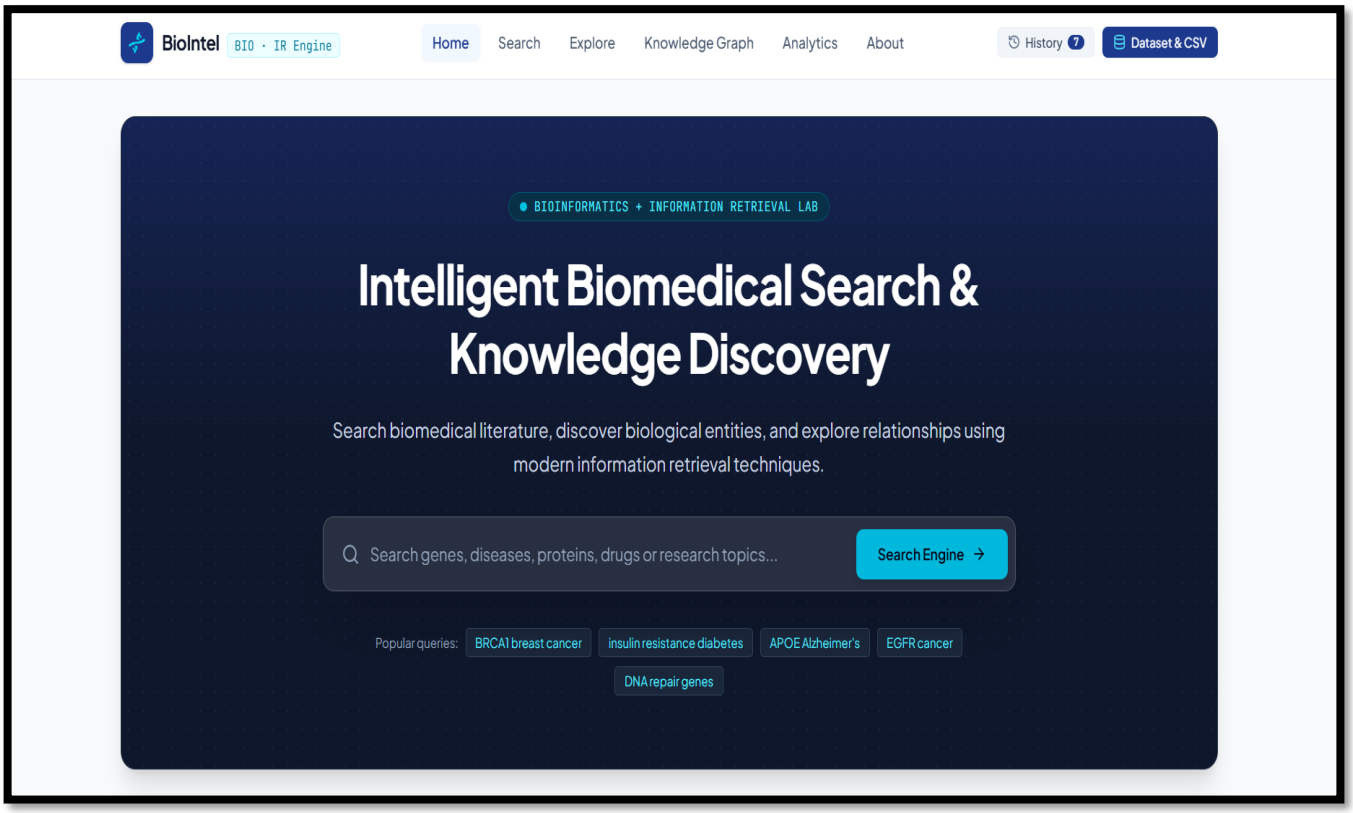
To keep the report similar to the sample report, you can add your own screenshots or diagrams at these points:

- After Section 5: system architecture or workflow diagram.
- After Section 7: screenshot of the BioIntel home page or search interface.
- After the entity-extraction explanation: screenshot showing highlighted genes, diseases, proteins, and drugs.
- After the knowledge-graph feature: screenshot of the interactive biomedical knowledge graph.
- After the analytics feature: screenshot of the dashboard charts.

Link To Prototype: BioIntel

Outputs:


BioIntel
BIO · IR Engine
Home
Search
Explore
Knowledge Graph
Analytics
About
History 7
Dataset & CSV
insulin resistance diabetes
Search
Suggestions:
BRCA1 breast cancer
insulin resistance diabetes
APOE Alzheimer's
EGFR cancer
DNA repair genes
Detected Query Entities:
insulin
diabetes
Entity:
Gene
Year:
2023
Ranking Model:
TF-IDF Cosine
Sort:
Oldest First
10 biomedical results retrieved for "insulin resistance diabetes"
Latency: 5.1 ms · Engine: TF-IDF
Insulin Resistance, Adipokines, and Metabolic Dysfunction in Type 2 Diabetes Pathogenesis
Relevance: 100%
Kahn SE, Cooper ME, Del Prato S · The Lancet · 2023 · 10.1016/S0140-6736(23)00412-9
Type 2 diabetes arises from a progressive deterioration of pancreatic beta-cell insulin secretion coupled with peripheral insulin resistance in skeletal muscle, liver, and adipose tissue. Pro-inflammatory cytokines including TNF and IL6 disrupt insulin receptor substrate signalin...
Entities:
TNF
IL6
Type 2 Diabetes
Metformin
View Paper
Why this result?
ID #5 · Indexed in BioIntel


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BIO · IR Engine
Home
Search
Explore
Knowledge Graph
Analytics
About
History 7
Dataset & CSV
BIOINFORMATICS + INFORMATION RETRIEVAL LAB
Intelligent Biomedical Search & Knowledge Discovery
Search biomedical literature, discover biological entities, and explore relationships using modern information retrieval techniques.
Search genes, diseases, proteins, drugs or research topics...
Search Engine →
Popular queries:
BRCA1 breast cancer
insulin resistance diabetes
APOE Alzheimer's
EGFR cancer
DNA repair genes